

Two-Tier Visibility Conditioning of the Acceleration Command

A minimal lean projection on the real camera plane, plus a self-releasing descent ease

Soft–Precise–Landing

9 September 2026 (supersedes the 31 Aug 2026 joint-QP note)

Scope. Keep the marker’s *measured centre* inside the *real* camera image plane, with a fixed buffer from the field-of-view edge, by making the smallest useful change to the desired inertial acceleration ${}^{\mathcal{I}}\mathbf{a}_d$ an outer loop produces. For the cross marker the visibility target is a *single point* — the crossed-lines intersection — so no 4-corner containment and no marker-size term. The command reaches that point through two channels on different timescales, handled as two tiers rather than one fused problem:

- **Attitude (lean).** A body tilt rotates the FoV, so the projected centre shifts within one control step, depth-free. This is the **hard** constraint (Tier 1).
- **Descent rate.** $|\tilde{\mathbf{r}}| \approx |\text{offset}|/Z$ grows as Z shrinks, at a rate $\sim |\tilde{\mathbf{r}}|(-\dot{Z}/Z)$. A change in the vertical command only affects the centre two integrators later, so descent is a **soft, predictive** influence (Tier 2): if the centre will reach the buffer before the lateral loop can re-centre it, ease off the descent to buy that loop time.

The tilt→feature coupling is the **rotational part of the IBVS interaction matrix**, L_ω , evaluated at the measured point — exact, *depth-independent*, no de-rotation or virtual frame. The 31 Aug 2026 note posed one QP over ${}^{\mathcal{I}}\mathbf{a}_d$ jointly enforcing visibility *and* thrust deliverability with a_z a solved variable; §8 records why that was retired.

Notation

Symbol	Meaning
$\tilde{\mathbf{r}}, \tilde{r}_k$	<i>measured</i> camera-frame marker centre, tangent units (px/ f), per axis k
$\varphi = \frac{\mathcal{R}}{2f}(1 - b)$	FoV-edge tangent half-extent minus the buffer; $b = \text{CBF_BUFFER_FRAC} = 0.15$
$L_\omega(\tilde{\mathbf{r}})$	rotational interaction matrix $\begin{bmatrix} xy & -(1+x^2) \\ 1+y^2 & -xy \end{bmatrix}$ at $(x, y) = \tilde{\mathbf{r}}$ — <i>no depth</i>
$M = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$	lean→rotation-axis map (fixed 90°): a lean along $\mathbf{y}$ is a rotation about $M\mathbf{y}$
$L_e = -(L_\omega M)$	effective Jacobian $\partial\tilde{\mathbf{r}}/\partial\mathbf{y}$. Sign negated vs the old $L_\omega M$ (§3)
$P = R_z(90^\circ)R_z(-\psi_b)$	inertial x - $y \rightarrow$ image axes (sim camera mounted yaw+90° on the down-pitch)
$\mathcal{I}\mathbf{a}_d$	desired inertial <i>specific thrust</i> , NED. Hover = $-g\mathbf{e}_3$; $\ \cdot\ $ is thrust/mass. Impl. I_a
$a_z = \mathcal{I}a_{d,z} $	vertical magnitude — a fixed input to Tier 1, never solved
$\mathbf{y}$	commanded lean $P(\mathcal{I}\mathbf{a}_{d,xy}/a_z)$, image axes; $\mathbf{y}_d$ from $\mathcal{I}\mathbf{a}_d^0$; $\mathbf{y}_{\text{curr}} = P(-R_{[2,3]}/R_{33})$ from the realized attitude
$\tau, \mathbf{d}$	drift look-ahead (s); measured centre drift rate (tangent/s, gyro-stripped). $\tau = 0$ for a stationary target
A_{cap}, g	vehicle max thrust-to-mass ratio; gravity
$g_z \in [g_{\min}, 1]$	Tier-2 descent-ease scale (§4); $g_{\min} = \text{CBF_GMIN} = 0.2$
T_{react}	lateral-loop reaction timescale, $\text{CBF_TREAT} = 1.5 \text{ s}$

1 Barrier on the real camera plane

Per image axis k ,

$$\boxed{h_k = \varphi_k - |\tilde{r}_k|}, \quad h_k \geq 0 \iff \text{the centre is on the sensor on axis } k. \quad (1)$$

$\tilde{\mathbf{r}}$ is the *measured* centre — nothing is de-rotated. φ is a fixed camera constant. The buffer b absorbs, in one number, the intended edge margin, the one-cycle linearization residual of (2), and inner-loop attitude-tracking lag.

2 Tilt→feature coupling, in acceleration units

A camera rotation moves the centre by L_ω , a standard *depth-independent* object at the measured point: $\dot{\tilde{\mathbf{r}}} = L_\omega \boldsymbol{\omega}_{\text{rp}} + \mathbf{d}$. The quantity a lateral-acceleration command sets is not a rotation rate but the underactuated quadrotor's *lean* $\mathbf{y}$; producing a steady lean requires a rotation about the perpendicular axis, $\boldsymbol{\omega}_{\text{rp}} = M\mathbf{y}$. Hence the feature response to the lean is $L_\omega M \mathbf{y}$, and to the lateral

acceleration directly

$$\tilde{\mathbf{r}}(\mathcal{I}\mathbf{a}_{d,xy}) \approx \tilde{\mathbf{r}} + L_e(\mathbf{y} - \mathbf{y}_{\text{curr}}) + \tau \mathbf{d}, \quad \mathbf{y} = P \frac{\mathcal{I}\mathbf{a}_{d,xy}}{a_z}, \quad (2)$$

with a_z a *given* (the outer loop's commanded value). Equation (2) is the local linearization, accurate over one control step; $\mathbf{y}_{\text{curr}}$ is read from the realized attitude, so it needs no rotation from acceleration units.

3 Tier 1 — the hard, this-cycle lean projection

$$\boxed{\mathcal{I}\mathbf{a}_d^*: \quad \mathbf{y}^* = \arg \min_{\mathbf{y}} \|\mathbf{y} - \mathbf{y}_d\|^2 \quad \text{s.t.} \quad |\tilde{\mathbf{r}} + L_e(\mathbf{y} - \mathbf{y}_{\text{curr}}) + \tau \mathbf{d}|_k \leq \varphi_k \quad (k = 1, 2)} \quad (3)$$

solved by a few alternating projections of $\mathbf{y}_d$ onto the two affine rows, then $\mathcal{I}\mathbf{a}_{d,xy}^* = a_z P^\top \mathbf{y}^*$ and $\mathcal{I}a_{d,z}^* = \mathcal{I}a_{d,z}$ **unchanged**.

- **Minimal intervention.** A row is touched only when it is *predicted* to breach. If $\mathbf{y}_d$ already keeps the centre in frame, $\mathbf{y}^* = \mathbf{y}_d$ exactly.
- **Outward-only.** Leaning *toward* the marker (or tangentially) reduces $|\tilde{\mathbf{r}}_k|$, so no row fires — a re-centring command is never clipped.
- **Depth-free, scale-free.** Every term is a camera intrinsic, an attitude, an image-space measurement, or a ratio/pass-through of the caller's own command.

Jacobian sign. $L_e = -(L_\omega M)$. The extra minus is not in the textbook forward-camera form; it was fixed against an *independent* pinhole + attitude camera model (`tools/validate_visibility_projection.py` check 1): for the *downward* camera the feature moves opposite to $L_\omega M \mathbf{y}$. The retired machinery used $L_\omega M$; its own offline validator carried a matching historical sign error, so the mistake was self-consistent and unseen.

Linearization envelope. (2) uses L_ω at the *measured* centre, fixed for the solve. It is $< 8\%$ off for a per-cycle lean *change* up to $\sim 15^\circ$ — the real regime at 50 Hz with a smooth outer loop — and the buffer b absorbs the residual (offline: 0/3742 true-sensor exits at $\leq 15^\circ/\text{cycle}$; 13/3762 at $\sim 24^\circ$, worst $+0.07$ tangent). It is *not* accurate for a one-shot $\sim 50^\circ$ correction; the every-cycle incremental nature of the correction and the lean cap below cover that.

Deliverability is separate. Thrust deliverability is *not* in (3). The caller applies it afterward: a lean cap $\|\mathbf{y}^*\| \leq \arccos(a_z/A_{\text{cap}})$ (the true deliverable tilt at the actual a_z ; θ_{cap} if $A_{\text{cap}} \leq g$), then a thrust-vector cap $\|\mathcal{I}\mathbf{a}_d\| \leq A_{\text{cap}}$. Folding these into the projection is what forced a_z to become a solved variable and created the coupled iteration of the old note; keeping them out keeps a_z a fixed input and the solve a plain feasibility projection.

4 Tier 2 — the soft, self-releasing descent ease

While the box binds it suppresses lateral authority the outer loop asked for. Pressing on at the full closing rate spends the descent budget before the lateral error is corrected. Tier 2 spends part of that budget on time, but keyed on a *measured* quantity, not on how much the box clipped:

$$\text{rem}_k = \varphi_k - |\tilde{r}_k|, \quad \text{close}_k = \max(\text{sgn}(\tilde{r}_k) \dot{\tilde{r}}_k, 0), \quad t_{\text{edge}} = \min_k \frac{\text{rem}_k}{\text{close}_k}, \quad (4)$$

$$\boxed{g_z = g_{\min} + (1 - g_{\min}) \text{clip}\left(\frac{t_{\text{edge}}}{T_{\text{react}}}, 0, 1\right), \quad \mathcal{I}a_{d,z}^{\star\star} = (\mathcal{I}a_{d,z}^{\star} + g)g_z - g} \quad (5)$$

applied only while descending ($\mathcal{I}a_{d,z}^{\star} + g > 0$) and low-passed so a_z does not step. t_{edge} is an image-distance over an image-speed, i.e. *seconds* — scale-free.

Why this is safe where the old descent relief was not. The retired CBF_AZ_COST_GAIN relief triggered on $\|\mathbf{y}_d - \mathbf{y}^{\star}\|$, a QP-internal norm that self-inflated as the relief’s own prior-iterate lift changed the a_z it was measured against, and it acted through $\min(\cdot, -g)$ which could *pin* a_z at hover (descent-freeze deadlock). Tier 2 instead: (i) triggers on an external measured signal ($|\tilde{r}|$ and its rate); (ii) runs *after* Tier 1, so it never fights the lean solve; (iii) is a *bounded scale* with $g_{\min} > 0$ — it can only slow a descent, and always leaves some; (iv) **self-releases**: the bought time shrinks $|\tilde{r}|$, so rem grows, t_{edge} grows, $g_z \rightarrow 1$.

5 Feasibility

(3) is infeasible iff $\varphi_k < 0$ for some k (the marker subtends the full FoV minus the buffer). Otherwise the alternating projection always converges. The large-tilt case where the linearized constraint and the true rotation homography diverge is not handled by a feasibility test but by the buffer plus the fact that the correction runs every 50 Hz cycle on a smooth command — a single cycle never demands a large lean change.

6 Assumptions

1. The coupling $L_e = -(L_{\omega}M)$ and its sign are calibrated (validator check 1 vs an independent pinhole model).
2. The conditioned command is realized: the inner attitude loop tracks the attitude built from $\mathbf{y}^{\star}$ on a timescale fast relative to the outer loop; residual lag is inside the buffer.
3. $\mathbf{d}$ is the translation part of the observed centre velocity, gyro-stripped; $\tau = 0$ for a stationary target (moving-target wiring is a separate step).
4. First-order L_{ω} over the cycle (envelope $\sim 15^\circ/\text{cycle}$; §3).
5. A_{cap}, g are fixed vehicle/physical constants, used directly.

7 Scale-freeness and depth-freeness

(1), (2), (3), (4)–(5) contain only the measured image point, $\varphi = \mathcal{R}/(2f)(1 - b)$, the body rate, the commanded acceleration and its measured image-space rate, A_{cap} , g , and fixed time constants. L_ω is depth-independent by construction; $t_{\text{edge}} = (\text{image distance})/(\text{image speed})$ is dimensionless in scale. The formulation is manifestly scale- and depth-free.

8 What this retired, and why

The 31 Aug 2026 joint QP over ${}^{\mathcal{I}}\mathbf{a}_d$ (visibility box + deliverability sphere, a_z solved), its two-phase δ / decode-fail geometry QP, the descent-rate relief (CBF_AZ_COST_GAIN), the frozen ρ_{fov} / $d_{\text{min,fov}}$ tilt-cone and its overflow/drift-off classification, and the ArUco 4-corner copy were all removed (cbf_visibility.py, cbf_visibility_aruco.py $\rightarrow$ Obsolete/). Grounds:

- The deliverability **sphere never bound** in flight (0.0% of gate frames; 81–97% of A_{cap}) — the post-step $\arccos(a_z/A_{\text{cap}})$ lean cap already covers it.
- With the sphere idle and the relief off, the joint projection is **algebraically the plain tilt projection** ($N_i(a_z){}^{\mathcal{I}}\mathbf{a}_{d,xy} \equiv L_\omega M \mathbf{y}$) — the a_z -in-the-loop iteration bought nothing.
- The descent relief **failed every SITL gate** in three forms; a clean IC2–5 A/B put CBF_AZ_COST_GAIN=0 ahead of =5 (tighter, softer, no regression).
- The ρ_{fov} cone was **deliberately frozen** (PLASMC_LFOV=0) and dormant.
- The two-phase δ apparatus fired **0.1% of frames** on the live cross-marker pipeline.

The only retained consumer of the old classification, the drift-off pull-back, is re-expressed as a per-axis increase of b on a persistent one-sided φ breach.

Validation of this design. tools/validate_visibility_projection.py — 10/10 against an independent pinhole + attitude oracle (barrier holds, minimal intervention, outward-free, conventions round-trip, descent-ease one-way / triggered-by-closing / self-releasing). SITL: IC2–5 $n = 5$ cross-marker A/B vs the retired machinery — both 20/20 land, 0 target-loss; this design pooled mean xy 0.14 m vs 0.24, precise 11/20 vs 8/20, max touchdown speed 1.47 vs 1.95 m/s. The median is a wash (on a clean approach the projection is a pure pass-through); the gain is entirely in the tail, where the retired joint-solve / relief / pull-back / cone-floor stack was found to thrash (intervening against the outer loop’s re-centring, chattering the tilt into a gain ratchet) on exactly the marginal off-centre approaches the gate stresses.